

Canonical Repaired Theorem Set Multi-Crop Ranking Reversal

Issue #2 theory audit deliverable

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1 Canonical model

On $(\Omega, \mathcal{F}, \mathbb{P})$ let $\mathcal{I} = \{1, \dots, n\}$ index crops. The ex-ante acreage decision is $x \in \mathbb{R}_+^n$. Per-hectare crop profit is $\pi_i = P_i Y_i - c_i$, portfolio profit is $\Pi(x) = \pi^\top x$, and loss is

$$L(x) = -\Pi(x), \quad \mu = \mathbb{E}[\pi].$$

The retained operational set is

$$X = \{x : \mathbf{1}^\top x \leq A, \ a^\top x \leq B, \ \ell \leq x \leq u, \ Gx \leq g\}.$$

Assume throughout that X is nonempty and compact and every π_i is integrable. For $\alpha \in (0, 1)$ define loss CVaR by

$$r_\alpha(x) = \inf_{v \in \mathbb{R}} \left\{ v + \frac{1}{1-\alpha} \mathbb{E}[(L(x) - v)_+] \right\}.$$

The canonical program and solution correspondence are

$$(P_\kappa) \quad \max_{x \in X} \{\mu^\top x : r_\alpha(x) \leq \kappa\}, \quad S(\kappa) = \arg \max(P_\kappa).$$

Marginal laws may be joined by a valid n -copula C_θ . The bivariate lower-tail coefficient, when its limit exists, is $\lambda_L = \lim_{q \downarrow 0} C(q, q)/q$; it is a diagnostic, not an ordering of general joint laws.

2 Reversal definitions

Let $s_i > s_j$ and let $\tau_x > 0$ be a declared numerical tolerance. A pair (i, j) is a possible reversal if some $x \in S(\kappa)$ has $x_j \geq x_i + \tau_x$, a universal reversal if every such x does, and a selected reversal if a declared selection $x^\sigma \in S(\kappa)$ does. A strong reversal has $x_i = 0 < x_j$. A top-rank reversal compares a highest-score crop with a lower-score crop. These labels are not interchangeable.

3 Results and proofs

Theorem 1 (CT1: existence, convexity, and finite LP). *r_α is finite, continuous, and convex on X . If (P_κ) is feasible, $S(\kappa)$ is nonempty and compact. Under finite scenarios π_s with weights $w_s > 0$, $\sum_s w_s = 1$, its risk constraint is represented exactly by*

$$v + \frac{1}{1-\alpha} \sum_s w_s q_s \leq \kappa, \quad q_s \geq -\pi_s^\top x - v, \quad q_s \geq 0, \quad v \in \mathbb{R}.$$

Proof. Each $x \mapsto -\pi(\omega)^\top x$ is affine. The positive part and expectation preserve convexity, and minimization in the Rockafellar–Uryasev representation gives the convex CVaR functional. Integrability and compact X give finiteness and continuity. Its sublevel set intersected with X is compact; a continuous linear objective attains its maximum there. For finite support, replacing every positive part by its epigraph variable q_s gives the displayed inequalities, and minimizing over q_s and free v recovers the definition exactly. $\square$

Theorem 2 (CT2: ordinal ranking insufficiency). *A strict weak order induced by suitability scores does not identify a cardinal acreage vector or an optimal allocation without an additional mapping from scores to payoffs and constraints.*

Proof. Every strictly increasing transformation of the scores preserves their order. Infinitely many cardinal vectors and feasible sets are compatible with that same order, and any selected feasible vector can be made uniquely optimal by a suitable linear objective over a suitable compact polytope. Thus the ordinal information alone cannot identify acreage. $\square$

Proposition 3 (CT3: feasibility-forced universal reversal). *If $s_i > s_j$ and*

$$\sup_{x \in X} x_i + \tau_x \leq \inf_{x \in X} x_j,$$

then every feasible point, hence every optimizer, reverses (i, j) . The readily checked condition $u_i + \tau_x \leq \ell_j$ is sufficient.

Proof. For every $x \in X$, $x_i + \tau_x \leq \sup_X x_i + \tau_x \leq \inf_X x_j \leq x_j$. Restriction to $S(\kappa) \subseteq X$ preserves the inequality. $\square$

Theorem 4 (CT4: exact risk-slack identity). *Let $S_0 = \arg \max_{x \in X} \mu^\top x$. Then*

$$S(\kappa) = S_0 \cap \{x : r_\alpha(x) \leq \kappa\}$$

whenever the right side is nonempty. Consequently all unconstrained optima remain optimal if all are risk-feasible; a single slack optimum does not imply equality of optimal sets under multiplicity.

Proof. No risk-feasible point can exceed the unrestricted optimum value. Every risk-feasible member of S_0 attains that value, while any risk-constrained optimizer attaining a smaller value would be dominated by such a member. Hence the two sets coincide. $\square$

Theorem 5 (CT5: complete KKT and finite tail weights). *Assume a convex constraint qualification. A feasible x^* is optimal if and only if there exist nonnegative multipliers $\eta, \beta, \rho, \gamma, \underline{\nu}, \bar{\nu}$ and $d \in \partial r_\alpha(x^*)$ satisfying*

$$0 = -\mu + \eta \mathbf{1} + \beta a + G^\top \gamma - \underline{\nu} + \bar{\nu} + \rho d$$

together with primal feasibility and complementary slackness for land, budget, every row of G , both bound systems, and risk. Under finite scenarios one may take

$$d = -\sum_s \xi_s \pi_s, \quad 0 \leq \xi_s \leq \frac{w_s}{1-\alpha}, \quad \sum_s \xi_s = 1,$$

with upper-tail saturation and VaR-atom splitting dictated by LP complementarity.

Proof. Apply convex KKT to minimization of $-\mu^\top x$ with all inequalities written in nonpositive form. The displayed stationarity is the subdifferential inclusion after collecting their normals; KKT necessity follows from the constraint qualification and sufficiency from convexity. For finite support, dualizing the excess-loss epigraph bounds yields the box constraints on ξ_s , while stationarity in the free variable v gives $\sum_s \xi_s = 1$. Stationarity in x yields the displayed subgradient, and complementarity assigns full weight above VaR, zero below, and admissible fractional weight at atoms. $\square$

Proposition 6 (CT6: risk-adjusted feasible displacement). *Let x be feasible and let $h \neq 0$ satisfy $x + th \in X$ for all small $t \geq 0$. Suppose $\mu^\top h > 0$ and either (i) $r_\alpha(x) < \kappa$, (ii) $r_\alpha(x) = \kappa$ and $r'_\alpha(x; h) < 0$, or (iii) direct evaluation establishes $r_\alpha(x + th) \leq \kappa$ for some small $t > 0$. Then x is not optimal.*

Proof. In case (i), continuity makes every sufficiently small step risk-feasible. In case (ii), directional differentiability gives $r_\alpha(x + th) = r_\alpha(x) + tr'_\alpha(x; h) + o(t) < \kappa$ for small positive t . Case (iii) supplies a feasible step directly. In all cases the objective rises by $t\mu^\top h > 0$. Strict negativity in (ii) cannot be weakened to zero: an active boundary locally shaped as $t^2 \leq 0$ has zero derivative but permits no positive step. $\square$

Theorem 7 (CT7: restricted dependence ordering). *Fix the marginals and a named copula family. If for $\theta_1 \preceq \theta_2$ and every x in a declared domain $D \subseteq X$*

$$L_{\theta_1}(x) \preceq_{\text{cx}} L_{\theta_2}(x),$$

then $r_{\alpha, \theta_1}(x) \leq r_{\alpha, \theta_2}(x)$ on D . Consequently the higher-dependence risk-feasible set within D is a subset of the lower-dependence one.

Proof. For each v , $z \mapsto (z - v)_+$ is convex. Convex order therefore orders its expectations. Add v , scale by $(1 - \alpha)^{-1}$, and take the infimum over v to obtain the CVaR inequality. Sublevel-set inclusion is immediate. No claim follows from λ_L alone. $\square$

Theorem 8 (CT8: reversal regions and crossing sets). *For a declared selection rule σ , define $\Delta_{ij}^\sigma(\theta) = x_j^\sigma(\theta) - x_i^\sigma(\theta)$ and*

$$\mathcal{R}_{ij}^\sigma = \{\theta : \Delta_{ij}^\sigma(\theta) \geq \tau_x\}, \quad \mathcal{C}_{ij}^\sigma = \{\theta : \Delta_{ij}^\sigma(\theta) = 0\}.$$

Without continuity, uniqueness, strict monotonicity, endpoint crossing, and active-set regularity, the crossing set may be empty, a singleton, multiple points, or an interval.

Proof. The set definitions are exact. Constant positive and constant negative gaps give empty crossing sets; an affine sign-changing gap gives one point; a polynomial with several roots gives several; and a function vanishing on an interval gives an interval. Optimization correspondences can realize discontinuity and multiplicity, so none of the omitted properties follows from the canonical program. $\square$

Proposition 9 (CT9: pseudo-diversification is diagnostic). *For predeclared thresholds ρ_0, λ_0 , the label $D_{\rho, \lambda} = \{(i, j) : |\rho_{ij}| \leq \rho_0, \lambda_{L, ij} \geq \lambda_0\}$ is descriptive only. Membership implies neither optimal inclusion or exclusion nor an ordering of portfolio CVaR or welfare.*

Proof. Optimal inclusion also depends on means, all constraints, the full joint law, and other crops. Raising a flagged crop's mean sufficiently can force inclusion; an upper bound or low mean can force exclusion independently of the flag. Equal scalar diagnostics can coexist with distinct joint laws and CVaR. Therefore no asserted optimization conclusion is logically entailed. $\square$

Theorem 10 (CT10: information actionability and weak flexibility). *Let a signal Z arrive before a recourse action is selected. If ignoring Z is admissible, posterior optimization has value no lower than prior optimization. Its value of information is zero when one common policy is optimal for almost every posterior problem. If $\mathcal{A}_{f_1}(z) \subseteq \mathcal{A}_{f_2}(z)$ almost surely, optimized value is weakly increasing in flexibility. Strict increase and supermodular information–flexibility complementarity do not follow without additional structure.*

Proof. The posterior decision maker can copy the prior policy, so optimization over signal-contingent policies weakly improves value. A common posterior optimizer attains both values, proving zero value. Maximization over a superset cannot lower value, proving weak flexibility monotonicity. Equality occurs whenever the added actions are never optimal, refuting strictness; substitutes between information and recourse refute general supermodularity. $\square$

4 Transition and evidence boundary

The Draft’s multi-crop architecture is retained. The false general ranking-reversal equivalence is replaced by CT3, CT5, and CT6; global tail-dependence monotonicity by conditional CT7; a unique threshold by CT8; the pseudo-diversification proposition by CT9; and strict information–flexibility complementarity by CT10. Every Draft result R01–R31 is dispositioned in the transition registry. All teacher-Draft numbers remain illustrative and are inadmissible as paper evidence until their separate simulation or empirical contracts are executed.

Method sources

The loss-CVaR representation and finite-scenario LP follow the full-text method foundations of Rockafellar and Uryasev (2000, 2002). Dependence language is restricted to named, verified families, consistent with the family-specific analysis of Ansari and Rockel (2024). Exact source URLs, DOIs, verification dates, roles, and limitations are recorded in the literature registry and method-source verification note.